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# CRITICAL HEAT FLUX PREDICTION AND UNCERTAINTY QUANTIFICATION WITH BAYESIAN OPTIMIZED DEEP ENSEMBLE

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## ABSTRACT

*Understanding Critical Heat Flux (CHF) is crucial for the safe operation of water-cooled nuclear reactors. Advancements in machine learning (ML) offer new opportunities to improve CHF predictions. The US Nuclear Regulatory Commission (US-NRC) released the NRC CHF database, which contains approximately 25,000 data points. These data are noisy and influenced by various sources of uncertainty, as they originate from multiple experiments. To understand the limitations of any neural network applied to this data, it is crucial to estimate the uncertainty associated with the inherent noise, known as aleatoric uncertainty, as well as the epistemic uncertainty, which arises from the model's limited knowledge.*

*To generate data-driven models with Uncertainty Quantification (UQ) capability, we applied the Deep Ensemble (DE) method combined with Bayesian Optimization (BO), named BODE, to optimize an ensemble of deep neural networks (DNNs) trained on the NRC CHF database, focusing on estimating both aleatoric and epistemic uncertainties. The approach improves predictive accuracy and uncertainty quantification. The optimized model exhibited improved predictive accuracy and reduced uncertainties when compared to the baseline model. The results of the BODE, and a comparison of the BODE and baseline ensemble models are presented, focusing on key metrics such as RMSPE, MAPE, and $EQ^2$.*

**Keywords: Critical heat flux, Neural network, Uncertainty quantification, Deep ensemble, Bayesian optimization**

## NOMENCLATURE

*Roman letters*

| | |
|---|---|
| $y$ | Target value (CHF) [W m$^{-2}$] |
| $k$ | Covariance function (kernel) of Gaussian Process |
| $\ell$ | Length scale parameter in RBF kernel |
| $i$ | Index of individual model in the ensemble |
| $j$ | Optimization iteration index |
| $M$ | Number of ensemble members |
| $n$ | Data point index |

*Greek letters*

| | |
|---|---|
| $\mu$ | Predicted mean [W m$^{-2}$] |
| $\sigma^2$ | Predicted variance [W$^2$ m$^{-4}$] |
| $\sigma_{\text{aleatoric}}$ | Aleatoric uncertainty [W$^2$ m$^{-4}$] |
| $\sigma_{\text{epistemic}}$ | Epistemic uncertainty [W$^2$ m$^{-4}$] |
| $\sigma_{\text{total}}$ | Total uncertainty [W$^2$ m$^{-4}$] |
| $\theta$ | Model hyperparameters |

## 1. INTRODUCTION

Understanding Critical Heat Flux (CHF) is essential for safe design principles in water-cooled nuclear reactors. CHF indicates the moment when heat transfer from a heat source's surface abruptly diminishes, causing a rapid rise in wall temperature and leading to dryout in boiling water reactors (BWRs) and Departure from Nucleate Boiling (DNB) in pressurized water reactors (PWRs) [1]. Due to the complex interaction of local thermal and fluid dynamics, predicting CHF analytically is challenging and often relies on specialized correlations or look-up tables derived from empirical data [2]. Over the past 60 years, more than a thousand correlations have been proposed to predict CHF values [3].

Recent advancements in data-driven models and Machine Learning (ML) have opened up new possibilities for improving the prediction of CHF [4, 5]. Further development, including this work, is ongoing through the Nuclear Energy Agency (NEA) Expert Group on Reactor Systems Multi-Physics (EGMUP) initiative to establish a benchmark for AI/ML applications [6]. This was motivated by the US Nuclear Regulatory Commission (US-NRC), which developed the Groeneveld CHF lookup table [2] and published a digital form known as the NRC CHF database [7], containing around 25,000 data points. The benchmark activities include leverging the CHF database to train ML models that improve the predictions of older models, and conducting verification, Validation and Uncertainty Quantification (VVUQ) for the

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developed ML models.

The experimental data are well-known for being noisy, with various sources contributing to this noise. The CHF database is derived from 59 experiments, making it crucial to estimate the associated uncertainty. Aleatoric uncertainty, which is related to data noise, is inherent to the data, while epistemic uncertainty arises from the model's lack of knowledge in representing the data. Aleatoric uncertainty is irreducible, but some models may overestimate it, whereas epistemic uncertainty can be reduced.

On the other hand, Optimizing the neural network is crucial to measure aleatoric uncertainty accurately, minimize epistemic uncertainty, and enhance prediction accuracy. Various Uncertainty Quantification (UQ) methods have been developed, each tailored with strengths and limitations based on application needs. Bayesian Neural Networks (BNNs) leverage Bayesian inference to provide uncertainty estimates but are often limited by their high computational demands and scalability challenges [8]. Monte Carlo (MC) dropout, a faster alternative, introduces dropout layers during prediction, yet its performance may fall short in complex neural networks [9]. In contrast, Deep Ensemble (DE) offers a scalable solution with strong predictive accuracy and reliable UQ capabilities [10]. DE methods have been proposed to improve generalization, mitigate overfitting, and quantify both aleatoric and epistemic uncertainties [11, 12].

A Neural Network (NN) depends on a variety of hyperparameters, including the number of layers, activation function, and learning rate [13]. Different configurations can significantly impact model performance. Hyperparameter optimization aims to identify the optimal hyperparameters that yield a highly accurate model within a reasonable time [14]. Two primary approaches exist: model-free optimization, which does not rely on surrogate or predictive models and includes methods like grid search, random search, and population-based algorithms (e.g., genetic algorithms, evolutionary algorithms, particle swarm optimization) [15], and model-based optimization, where techniques such as Bayesian Optimization (BO) use a surrogate model to guide the search for optimal hyperparameters [16].

In previous work, we combined BO with DE, a method we refer to as BODE, to enhance the prediction and UQ [17]. The BODE was applied to a case study for the prediction of eddy viscosity using a Densely Connected Convolutional Neural Network (DCNN) and trained on CFD data [17] [18]. In that work, Gaussian noise was added to the CFD data to asses the performance of BODE on estimating the noise. The BODE was able to enhance the prediction and estimate data noise accurately.

Because of our success in the previous work, and understanding the importance of CHF for nuclear safety, in this work, we apply the BODE to an ennsemble of DNNs trained on the NRC CHF database. The primary objective is to improve predictions using five input features by applying BO for hyperparameter tuning to enhance the prediction and UQ. The optimization is conducted on a validation dataset that is not part of the training data to avoid overfitting, and the model performance is evaluated on a test dataset that wasn't used either in training or optimization.

The rest of the paper is organized as follows: Section 2 introduces the methodology; Section 3 discusses the obtained BODE results; Section 4 summarizes this work.

## 2. METHODOLOGY

An ensemble of DNNs is designed to predict the CHF, where each DNN uses five input features, configured with various hyperparameters, and optimized with BO. The hyperparameters include the learning rate (which determines the step size for updating weight coefficients during gradient descent), the dropout rate (the likelihood of randomly dropping neurons during training to mitigate overfitting), the number of hidden layers (the intermediate layers between the input and output layers), the number of hidden units (the count of neurons in each hidden layer), the activation function (used to introduce non-linearity after each hidden layer), and weight decay (a regularization method that discourages large weight coefficients to reduce overfitting).

### 2.1 Deep Ensemble

An ensemble of 10 neural networks is employed mainly to conduct UQ, also, it helps to enhance the model's generalization performance and mitigate overfitting. Instead of focusing solely on target value prediction, the DE method extends its utility by offering estimates of predictive uncertainty through a collection of neural networks [19]. The ensemble is constructed by selecting 10 models from a pool of optimized candidates, all determined through BO.

Within the DE framework, predictions are treated as samples drawn from a Gaussian distribution, where each output consists of both a mean and a variance. The variance corresponds to aleatoric uncertainty while epistemic uncertainty is obtained by combining predictions across multiple models [20].

Aleatoric uncertainty, which arises from intrinsic randomness and noise within the data, is irreducible; while it can be accurately quantified, it cannot be eliminated. Conversely, epistemic uncertainty, arising from insufficient knowledge of the true model, can be reduced by gathering additional data or refining the model architecture [21].

When the data is assumed to follow a Gaussian distribution, least squares regression can be interpreted as a maximum likelihood estimation process, where predictions encompass both a mean and a variance, as described in Equation(1).

$$y(x) \sim N(\mu(x), \sigma^2(x)) \tag{1}$$

To achieve this, the negative log-likelihood of a Gaussian distribution is utilized as the loss function. Equation (2) expresses the probability of predicting a target value $y_n$ based on an input vector $x_n$, within a model parameterized by $\theta$. In this model, $\mu(x_n)$ represents the predicted mean, and $\sigma^2(x_n)$ signifies the predicted variance [22].

$$p_\theta(y_n|x_n) = \frac{1}{\sqrt{2\pi\sigma^2(x_n)}} \exp\left(\frac{-(y_n - \mu(x_n))^2}{2\sigma^2(x_n)}\right) \tag{2}$$

Maximizing the likelihood is equivalent to minimizing the negative log-likelihood. The expression for the negative log-likelihood is provided in Equation (3).

$$-\log p_\theta(y_n|x_n) = \frac{\log \sigma^2(x_n)}{2} + \frac{(y_n - \mu(x_n))^2}{2\sigma^2(x_n)} \tag{3}$$


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In this framework, $\sigma^2(x_n)$ captures the aleatoric variance, reflecting the Gaussian distribution inherent to the data. Estimating epistemic uncertainty, however, relates to the model's ability to generalize and thus requires multiple models. To address this, an ensemble of models $M$ is trained, with the final prediction being the average of all model outputs, as shown in Equation (4).

$$\mu(x) = \frac{1}{M}\sum_{i=1}^{M}\mu_i(x) \tag{4}$$

The total uncertainty $\sigma^2(x)$ is calculated by combining the individual model uncertainties $\sigma_i^2(x)$ and predictions $\mu_i(x)$, in relation to the ensemble prediction $\mu(x)$, as shown in Equation (5).

$$\sigma^2(x) = \frac{1}{M}\sum_{i=1}^{M}\left(\sigma_i^2(x) + \mu_i^2(x)\right) - \mu^2(x) \tag{5}$$

This overall uncertainty can be separated into aleatoric and epistemic components, as presented below [19, 23]:

$$\sigma^2_{\text{aleatoric}}(x) = \frac{1}{M}\sum_{i=1}^{M}\sigma_i^2(x) \tag{6}$$

$$\sigma^2_{\text{epistemic}}(x) = \frac{1}{M}\sum_{i=1}^{M}\left(\mu_i^2(x) - \mu^2(x)\right) \tag{7}$$

## 2.2 Optimization Strategy

The DNNs are optimized with BO to accurately estimate the aleatoric uncertainty, minimize the epistemic unceratinty, and enhance the prediction accuracy. This study utilizes the *Ax-platform* [24], an open-source optimization framework by Meta built on PyTorch [25], to perform the hyperparameter optimization. The Root Mean Square Error (RMSE), as defined in Equation (8), serves as the objective function, with the hyperparameters of the DCNN as its independent variables. In this context, the mean $\mu$ is the same as that in Equation (4). Although the Negative Log-Likelihood (NLL) loss function is a possible alternative to RMSE, RMSE is preferred here for its computational simplicity, as the variance terms in NLL can introduce competing effects that complicate optimization.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \mu_i)^2} \tag{8}$$

**TABLE 1: THE BAYESIAN OPTIMIZATION HYPERPARAMETERS SEARCH SPACE.**

| Hyperparameter | Range/Values |
|---|---|
| Learning rate | [0.0001 – 0.01] |
| Weight decay | [0.0001 – 0.01] |
| Drop rate | [0 – 0.3] |
| Number of hidden layers | {6, 7} |
| Number of hidden units | {8, 16, 24, 32, 48, 64, 96} |
| Activation function | {ReLU, LeakyRelu, Gelu, Selu, Elu, Softplus} |

Six hyperparameters are optimized: learning rate, weight decay, drop rate, batch size, the number of hidden layers, the number of hidden units, and the activation function. Each optimization model is initialized with unique values for these hyperparameters. Table 1 provides an overview of the defined search space for each parameter.

The data is divided into three datasets: training, validation, and testing datasets. The test dataset is not used during training or optimization. The Bayesian optimization is conducted to minimize the prediction loss on the validation dataset. The training, validation, and test datasets constitute 72%, 18%, and 10% of the NRC CHF database, respectively.

The optimization process consists of two main phases: the Sobol sequence and Bayesian Optimization (BO).

1. **Sobol sequence**: A quasi-random deterministic sequence that ensures uniform coverage of the multidimensional search space. It is utilized to generate various configurations of the hyperparameters, guaranteeing good coverage of the hyperparameter search space before initiating the BO step. It serves as a prior distribution for the BO.

2. **Bayesian Optimization**: An optimization strategy employed to minimize the RMSE of the validation dataset. Further details on how it operates are provided in Section 2.3.

Five optimization processes are intiated, and the best 10 performing models are chosen. Since the Sobol sequence is a deterministic sequence different initial random hyperparameter setups are used to tackle the hyperparameter search space from different points. This initial random hyperparameter setup acts like a seed to the Sobol sequence. Starting from the seed, the Sobol sequence iteratively generates new sets of hyperparameters. The model is trained using the generated hyperparameters in sequence, with the RMSE of the validation dataset evaluated for each model. This iterative process builds the Gaussian Process (GP) prior distribution, enhancing the search for optimal hyperparameters based on observed RMSE outcomes.

## 2.3 Bayesian Optimization

BO starts after the completion of the Sobol sequence step. BO is a highly effective technique for optimizing black-box functions that are computationally expensive to evaluate. Its objective is to determine the input parameters that either maximize or minimize the target function.

BO relies on two key components: a surrogate model and an acquisition function. The surrogate model is a probabilistic representation of the target function, built using a limited number of sampled points, and it improves in accuracy as additional points are evaluated. Among the available surrogate models, Gaussian Processes are the most widely adopted for BO. In this work, BoTorch's *SingleTaskGP* is used as the surrogate model.

A GP is defined by a mean function $\mu(x)$ and a covariance function $k(x, x')$. The Root Mean Square Error (RMSE) is approximated through the GP, as described in Equation (9), with the Radial Basis Function (RBF) employed as the covariance function, shown in Equation (10). Furthermore, a constant mean

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**TABLE 2: BAYESIAN OPTIMIZATION (BO) STRATEGY**

| Step | Description |
|---|---|
| Initialization | Select 5 random configurations of the hyperparameters from the search space to start the optimization processes. |
| | **For loop (1) for** $i = 0, 1, \ldots, 5$ |
| Begin Optimization | Start optimization for each of the optimization processes. |
| | **For loop (2) for** $j = 1, 2, \ldots, 10$ |
| Prior Construction | Use the Sobol sequence to establish the initial prior distribution for the Gaussian Process (GP) over the first 10 iterations. |
| | **End for loop (2)** |
| | **For loop (3) for** $j = 10, 11, \ldots, 40$ |
| Sampling | Choose a new point $x_{n+1}$ using the acquisition function. |
| Model Update | Refresh the GP surrogate model with the new sample. |
| | **End for loop (3)** |
| | **End for loop (1)** |
| Optimization Outcome | Identify 10 hyperparameter configuration that achieves the lowest loss to build the ensemble from the 5 optimization process. |

function $\mu$ is used, ensuring that in the absence of data, the GP defaults to a value near this mean.

$$RMSE \sim \mathcal{GP}(\mu(x), k(x, x')) \tag{9}$$

$$k_{\text{RBF}}(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right) \tag{10}$$

In the BO process, the surrogate model is iteratively improved using an acquisition function, which determines the next point to evaluate. The acquisition function strategically balances *exploration*—sampling areas where the surrogate model's predictions carry high uncertainty—and *exploitation*, targeting regions where the model predicts low loss values.

Given a set of observed data points $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, a GP can estimate the function value at a new location $x^*$. This prediction results in a normal distribution characterized by a mean $\mu(x^*)$ and variance $\sigma^2(x^*)$. Initially, the GP is assigned a prior distribution over the possible objective functions, guided by preliminary observations generated using the Sobol sequence. As new data points are acquired, the GP model updates to form a posterior distribution, progressively enhancing its representation of the objective function by incorporating the newly gathered information.

Using the GP prior established from the Sobol observations, the acquisition function selects the next hyperparameter configuration to evaluate. For each new configuration, the model is trained, the RMSE for the optimization validation dataset is computed, and this new result is added to the surrogate model. The surrogate model is continually updated throughout the BO process. To construct the ensemble, this optimization process is repeated 5 times in parallel, with each run initialized using a distinct set of hyperparameters generated by the Sobol sequence. This ensures diversity among the prior Gaussian distributions in the GP surrogate models, leading to a diverse ensemble of models. The best ten performing models from the optimization processes are chosen as ensemble members. The entire optimization algorithm is summarized in Table 2.

The *qNoisyExpectedImprovement* acquisition function, defined in Equation (11), is utilized in this study. This function evaluates the expected improvement of candidate points compared to the best value observed to date. In this context, $Y$ denotes the predicted value at the new point, while $Y_{\text{baseline}}$ represents the value from the previously observed data.

$$q\text{NEI}(\mathcal{X}) = E\left[\min\left(\min(Y) - \min(Y_{\text{baseline}}), 0\right)\right] \tag{11}$$

## 2.4 Baseline Ensemble

The baseline ensemble serves as a reference to compare the performance of the BODE. It is constructed through manual iteration of the hyperparameters.

## 2.5 Data Preprocessing

The NRC CHF database includes seven input variables: pressure, temperature, heated length, tube diameter, mass flux, quality, and inlet subcooling. Most current models use only five input features, including pressure, heated length, tube diameter, mass flux, and quality, while the inlet subcooling and temperature are not included as they are considered redundant. The rationale is that redundancy reduces the model's generalization capability and increases overall complexity.

Data preprocessing steps include log transformation, and normalization. Figure 1 presents histograms of the original input features and the target CHF. After preprocessing, the data distributions are shown in Figure 2.


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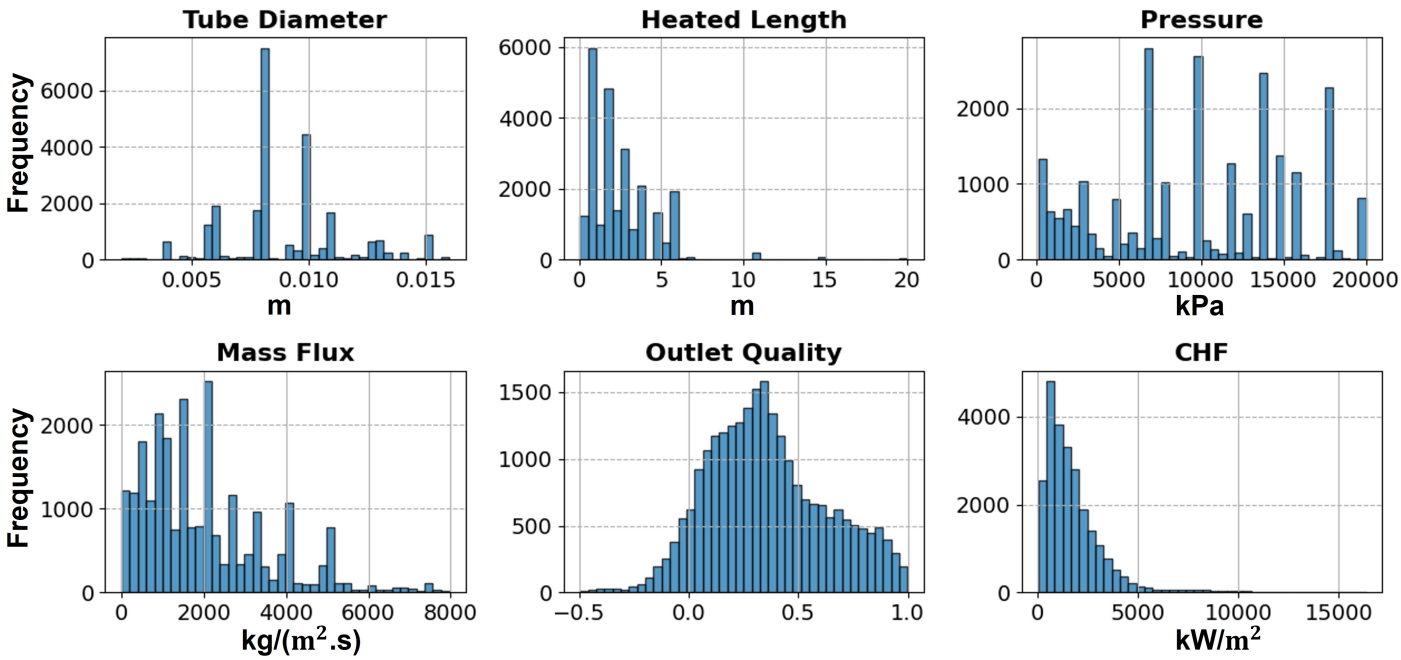


**FIGURE 1: HISTOGRAMS OF THE ORIGINAL INPUT FEATURES AND THE TARGET CHF.**

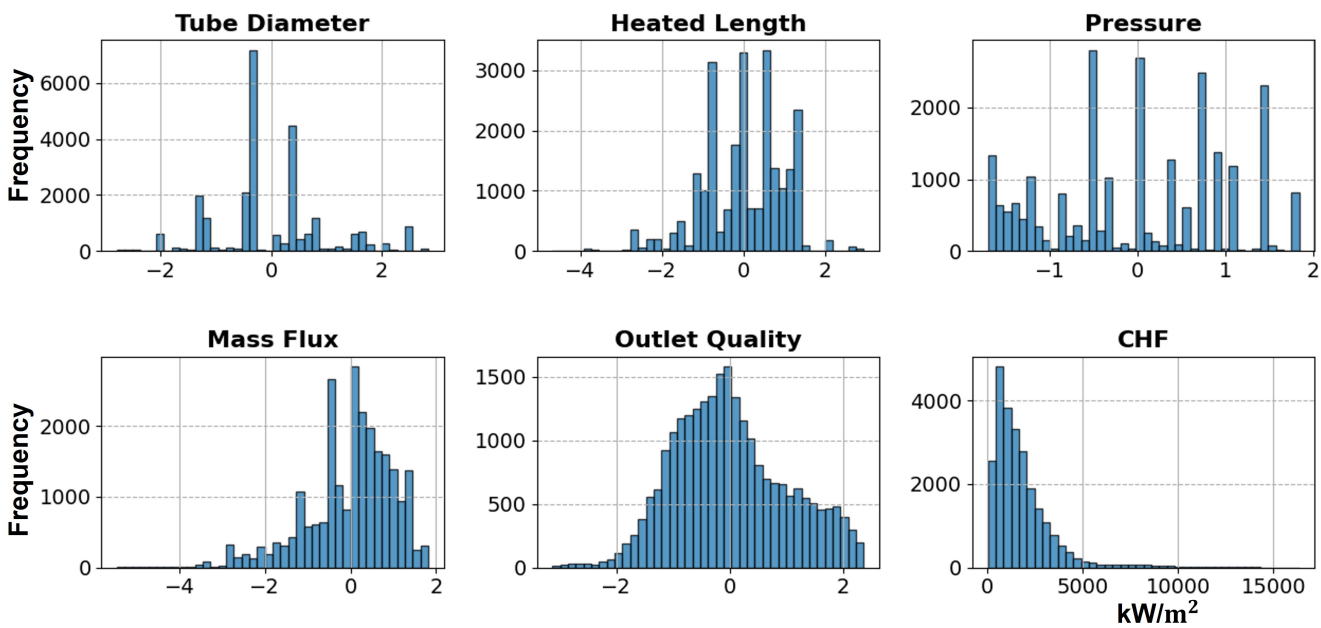


**FIGURE 2: HISTOGRAMS OF THE INPUT FEATURES AND THE TARGET CHF AFTER DATA PROCESSING.**

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## 3. RESULTS

Figure 3 illustrates the loss versus training iterations for one of the ensemble members, indicating the model's convergence behavior during training. The plot displays the RMSE values for both the training and validation datasets. The close similarity in RMSE values between the training and validation datasets is attributed to using the validation dataset for the optimization. This alignment ensures comparable predictive accuracy for both the training and validation data. The primary objective of the optimization is to minimize the RMSE on the validation dataset. By focusing on the validation dataset rather than the training dataset, the risk of overfitting is reduced.

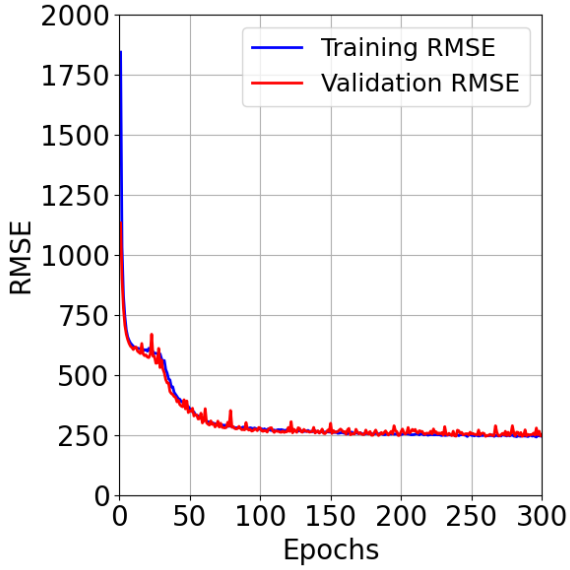


**FIGURE 3: LOSS VS. TRAINING ITERATION FOR ONE OF THE OPTIMIZED ENSEMBLE MEMBERS.**

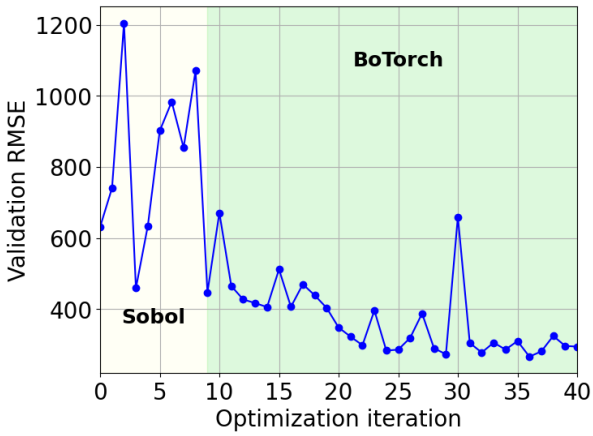


**FIGURE 4: THE OPTIMIZATION ITERATIONS IN SOBOL SEQUENCE AND BAYESIAN OPTIMIZATION.**

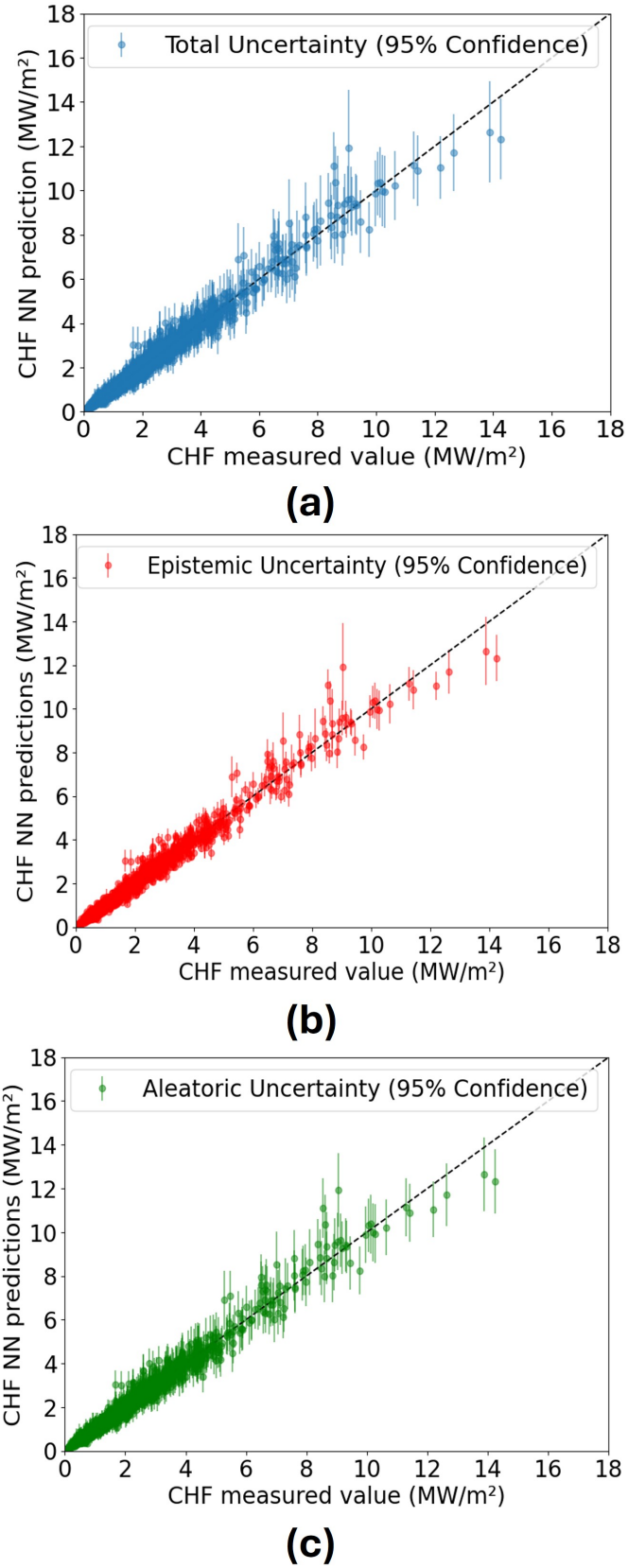


**FIGURE 5: BODE CHF PREDICTION VS. MEASUREMENT AND UNCERTAINTY QUANTIFICATION RESULTS FOR THE TESTING DATASET IN TERMS OF (A) TOTAL UNCERTAINTY, (B) EPISTEMIC UNCERTAINTY, (C) ALEATORIC UNCERTAINTY.**

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The optimization process consists of 41 iterations, a number determined heuristically. Initially, for the first 10 iterations, a Sobol sequence is used to establish a prior distribution for the GP surrogate model. Subsequently, BO is performed using BoTorch to optimize the models. During each iteration, the DNN model is trained on the training dataset and optimized on the validation dataset based on the RMSE. These RMSE values are then used to build and update the GP surrogate model. At each step of the BO process, the acquisition function selects the next hyperparameter configuration to sample, guided by the current state of the surrogate model.

Figure 4 illustrates the optimization iterations during the Sobol sequence and Bayesian optimization phases. The large fluctuations in the loss values during the Sobol stage are due to the high variability in hyperparameter setups, as the objective in this stage is to explore the hyperparameter search space rather than optimize the model. In the BO stage, the RMSE decreases over the iterations, demonstrating that the surrogate model increasingly captures the relationship between hyperparameters and the RMSE objective function with greater accuracy as more observations are added. The observed fluctuations are attributed to the exploration aspect of the acquisition function, which maintains a balance between exploration and exploitation during optimization.

Figure 5 depicts the NN predictions of CHF values compared to the measurements for a blind test dataset. The predictions are shown alongside the total, epistemic, and aleatoric uncertainties, each represented with 95% confidence intervals. The predictions align closely with the diagonal line, indicating good agreement between predictions and measurements. Figure 5.a illustrates the total uncertainty, which encompasses both epistemic (model-related) and aleatoric (data-related) uncertainties, as defined in Equation 5.

Total uncertainty increases for higher CHF values, as do both epistemic and aleatoric uncertainties, as shown in Figures 5.b and 5.c. The elevated epistemic uncertainty is most pronounced at extreme CHF values due to sparse data, while the higher aleatoric uncertainty at these extremes is attributed to greater noise in the measurements. The figure also shows that the epistemic uncertainty of some data points is very small (close to zero). On the other hand, the aleatoric uncertainty couldn't be close to zero.

Figure 6 shows histograms of the BODE CHF prediction-to-measurement ratios for the training, validation, and testing datasets when applying the BODE method. Figures 6.a, 6.b, and 6.c display the histograms for the training, validation, and test datasets, respectively. The results reveal a consistent alignment across all datasets, with the ratios centered around 1 and narrow distributions, indicating low variability in the predictions. The pronounced peak at 1 highlights a good agreement between the predicted and measured CHF values. The narrow distribution and consistent peak across all datasets demonstrate that the model generalizes effectively to unseen data, avoiding both underfitting and overfitting.

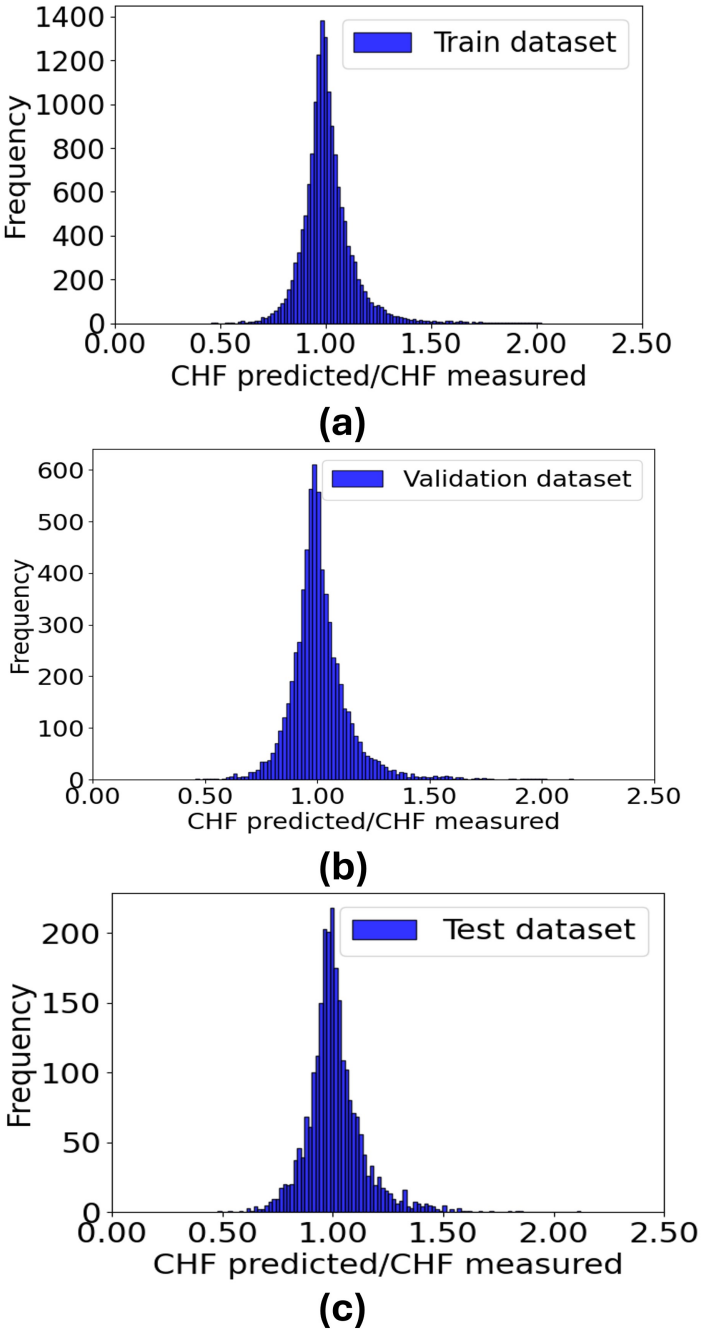


**FIGURE 6: BODE CHF PREDICTION OVER MEASUREMENT HISTOGRAMS FOR (A) TRAINING DATASET, (B) VALIDATION DATASET, (C) TESTING DATASET.**

Figure 7 illustrates the BODE prediction-to-measurement ratios plotted against various input features for the training, validation, and testing datasets. Each subplot corresponds to a specific input feature: tube diameter, heated lengthm outlet quality, Mass Flux, and pressure.

Across all datasets and input features, the predicted-to-measured ratio consistently centers around 1, indicating that the model's predictions align closely with the measured values. The

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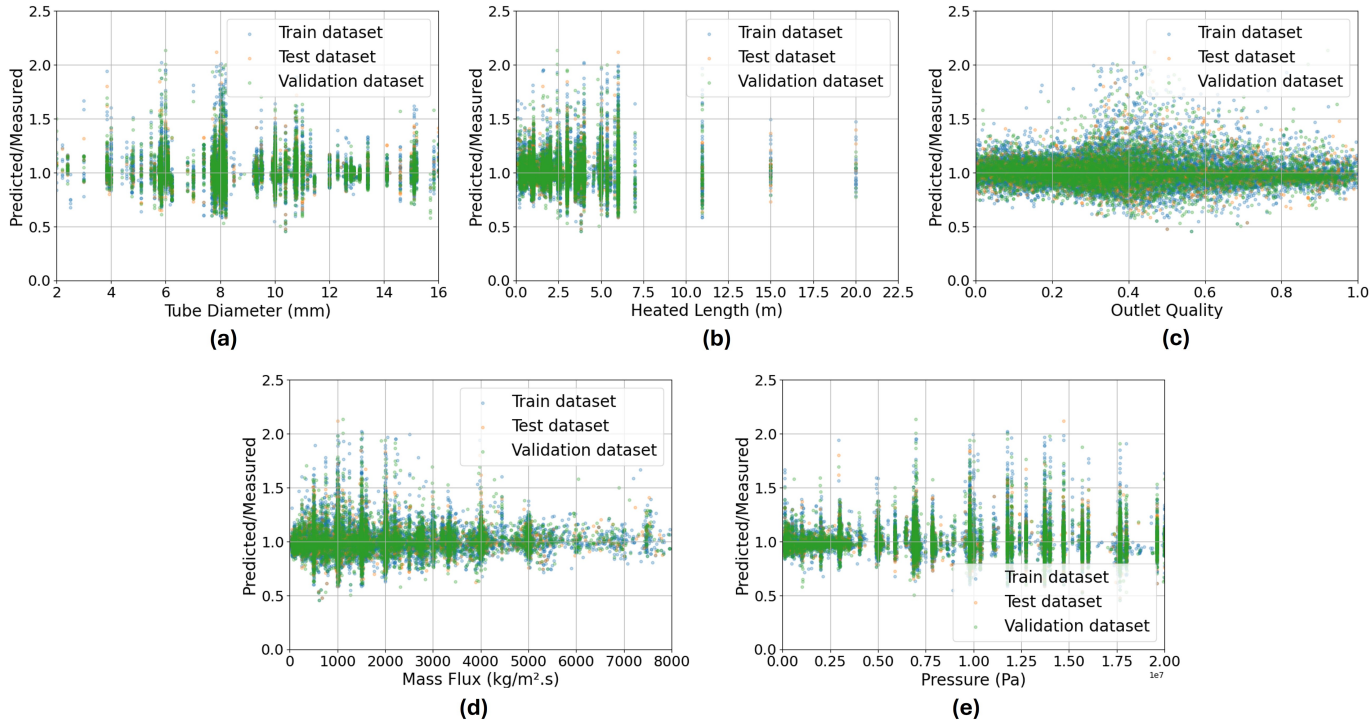


**FIGURE 7: BODE PREDICTION OVER MEASUREMENT AGAINST INPUT FEATURES: (A) TUBE DIAMETER, (B) HEATED LENGTH, (C) OUTLET QUALITY, (D) MASS FLUX, AND (E) PRESSURE.**

majority of the data points fall within the range of 0.8 to 1.2, with some data scattered beyond this interval, further demonstrating the model's reliability across various features.

The performance of the BODE model was evaluated using various error metrics, as shown in table 3. The table compares the Mean Prediction over Measurement (P/M), Standard Deviation of P/M, Root Mean Square Percentage Error (RMSPE), Mean Absolute Percentage Error (MAPE), and $EQ^2$ Error across the training, validation, and test datasets. For the mean P/M, both ensembles yield values close to 1, indicating good alignment between predictions and measurements. However, the BODE model achieves a lower standard deviation of P/M, highlighting a better consistency compared to the baseline ensemble. Moreover,

the BODE model demonstrates a reduction in RMSPE and MAPE, emphasizing its enhanced accuracy relative to measured values. The $EQ^2$ metric evaluates a model's ability to predict target values relative to data variability using the mean as a baseline. A $EQ^2$ value greater than 1 indicates poorer performance than using the data mean, while a value of 0 reflects perfect prediction accuracy. The BODE model consistently exhibits lower $EQ^2$ values than the baseline ensemble, demonstrating its improved predictive capabilities.

## 4. CONCLUSION

Accurate CHF predictions with reliable UQ are essential in nuclear reactor safety analysis, where decision-making relies heavily on the confidence in model outputs. In this study, we apply a novel approach that combines Bayesian Optimization (BO) with Deep Ensembles (DE), referred to as BODE, to enhance both predictive accuracy and uncertainty quantification (UQ) in Deep Neural Networks (DNNs) for Critical Heat Flux

**TABLE 3: ERROR METRICS FOR DIFFERENT DATASETS**

| Dataset | Method | Mean P/M | Std P/M | RMSPE [%] | MAPE [%] | $EQ^2$ Error |
|---|---|---|---|---|---|---|
| Training | BODE | 1.0058 | 0.1168 | 11.69 | 8.160 | 0.01987 |
| | Baseline | 0.9967 | 0.1204 | 12.05 | 8.472 | 0.02340 |
| Validation | BODE | 1.0067 | 0.1277 | 12.78 | 8.832 | 0.02113 |
| | Baseline | 0.9973 | 0.1290 | 12.90 | 9.010 | 0.02304 |
| Test | BODE | 1.0068 | 0.1276 | 12.78 | 8.849 | 0.02140 |
| | Baseline | 0.9984 | 0.1295 | 12.95 | 9.039 | 0.02373 |
| All | BODE | 1.0059 | 0.1168 | 11.69 | 8.159 | 0.01987 |
| | Baseline | 0.9969 | 0.1175 | 12.38 | 8.674 | 0.02336 |


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(CHF) prediction. Our approach addresses the limitations of traditional DE methods, which often overestimate aleatoric uncertainty and exhibit discrepancies between model predictions and actual data, especially when constructed by retraining the same neural network multiple times with randomly initialized weights. By optimizing the hyperparameters of the DNN through BO, we achieved improvements over a manually tuned baseline ensemble. Specifically, the BODE method effectively minimized epistemic uncertainty by selecting optimal hyperparameters, while more accurately estimating the aleatoric uncertainty inherent in the experimental CHF data.

Our results demonstrated that the BODE approach outperforms the baseline model in terms of prediction accuracy, achieving lower Root Mean Square Percentage Error (RMSPE) and Mean Absolute Percentage Error (MAPE) across training, validation, and test datasets. Moreover, the BODE method provided more reliable uncertainty estimates, accurately reflecting the data noise and the model's confidence in its predictions.

In the context of nuclear reactor design and operation, where accurate modeling of complex phenomena like CHF is critical for safety analyses, the BODE method offers a promising approach to enhance predictive capabilities while providing reliable UQ. This can contribute to better risk assessment, improved safety margins, and informed decision-making. While this study focused on CHF prediction using the NRC CHF database, the methodology is flexible and can be extended to other domains and neural network architectures.

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